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MILITARY SPECIFICATION

DRY CLEANING PRESSES AND FINISHING EQUIPMENT, COMMERCIAL (SHIPBOARD)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope This specification covers commercial dry cleaning presses and finishing equipment for Navy shipboard use

1.2 Classification Dry cleaning presses and finishing equipment shall be of the following types and classes as specified (see 6.2):

Type I - Presses, wool, utility.

Class B - Press, wool, utility, pneumatically operated
Class C - Press, wool, utility, legger, pneumatically operated, 2-lay, with square end

Type III - Board, sporting, steam unit.
Type V - Presses, wool, topper, pneumatic, automatic .

Class A - Press, wool, pants topper, pneumatic, automatic, 1-lay, without pleat clamps

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 5523, Department of the Navy Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications. standards. and handbooks, The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-F-320	- Fiberboard, Corrugated and Solid Sheet Stock (Container Grade), and Cut Shapes.

MILITARY

00-D-735	- Dry Cleaning Finishing Equipment. Miscellaneous.
MIL-E-917	- Electric Power Equipment, Basic Requirements (Naval Shipboard Use).
MIL-L-3153	- Laundry and Dry Cleaning Machinery and Equipment, Packaging of.
MIL-S-8660	- Silicone Compound, NATO Code Number S-736.
MIL-L-15719	- Lubricating Grease (High-Temperature, Electric Motor, Ball and Roller Bearings)
MIL-M-17060	- Motors, 60-Hertz, Alternating Current, Integral-Horsepower (Shipboard Use).
MIL-L-17331	- Lubricating Oil, Steam Turbine and Gear, Moderate Service.
MIL-L-19140	- Lumber and Plywood, Fire-Retardant Treated.
DOD-G-24508	- Grease, High Performance, Multipurpose. (Metric)

STANDARDS

MILITARY

MIL-STD-167-1	- Mechanical Vibrations of Shipboard Equipment (Type T - Environmental and Type II - Internally Excited).
MIL-STD-1399, Section 300	- Interface Standard for Shipboard Systems Electric Power, Alternating Current. (Metric)
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099,)

2.2 Non-Government Duplications . The following document(s) form a part of this document to the extent specified herein,. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

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AMERICAN IRON AND STEEL INSTITUTE (AISI)
Steel Products Manual.

(Application for copies should be addressed to the American Iron and Steel Institute, 350 Fifth Avenue, New York, NY 10017.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)
Boiler and Pressure Vessel Code.
Section VIII - Rules for Construction of Pressure Vessels.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)
Y14.5M - Dimensioning and Tolerancing.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERCIAN SOCIETY FOR TESTING PAD MATERIALS (ASTM)
A 36 - Standard Specification for Structural Steel. (DoD adopted)
A 47 - Standard Specification for Ferritic Malleable Iron Castings (DoD adopted)
A 109 - Standard Specification for Steel, Carbon, Cold-Rolled Strip, (DoD adopted)
A 240 - Standard Specification for Heat-Resisting Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels. (DoD adopted)
A 269 - Standard Specification for Seamless and Welded Austenitic Stainless Steel Tubing for General Service (DoD adopted)
A. 446 - Standard Specification for Steel Sheet, Zinc-Coated (Calvanized) by the Hot-Dip Process, Structural (Physical) Quality. (DoD adopted)
A 478 - Standard Specification for Chromium-Nickel Stainless and Heat-Resisting Steel Weaving Wire.
B 61 - Standard Specification for Steam or Valve Bronze Castings (DoD adopted)
B 62 - Standard Specification for Composition Bronze or Ounce Metal Castings. (DoD adopted)
B 75 - Standard Specification for Seamless Copper Tube. (DoD adopted)
B 127 - Standard Specification for Nickel-Copper Alloy (UNS N04400)plate, Sheet, and Strip,
B 209 - Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate, (Dot) adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)
IEEE 45 - IEEE Recommended Practice for Electric Installations on Shipboard. (DoD adopted)

(Application for copies should be addressed to the Institute of Electrical and Electronics Engineers, 345 East 47th Street, New York, NY 10017.)

UNDERWRITERS LABORATORIES INC. (UL)
UL 353 - Limit Controls.

(Application for copies should be addressed to the Underwriters Laboratories Inc, 333 Pfingsten Road, Northbrook, IL 60062.)

(Non-government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of Precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets or MS standards) , the text of this document takes precedence Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6 2), a sample shall be subjected to first article inspection (see 6.5) in accordance with 4.4.

3.2 Materials, Materials shall be as specified in 3,2.1 through 3.2.11.

3.2 1 Structure. Types I, III, and V shall be constructed for deck mounting and shall have steel frames, Strength members, lever systems, bearing supports, bases, columns, and similar items shall be constructed of malleable iron as specified in ASTM A 47 or steel as specified in ASTM A 36. Gray cast iron or other brittle materials shall not be used for structural parts Gray cast iron may be used for components such as gear reducers and bearing housings, which are not strength members unless manufactured as an integral part of the frame Those components shall satisfactorily complete the vibration testing specified in MIL-STD-167-1. The structure shall be finished, plated, or made to prevent corrosion in a salty (marine) atmosphere (see 4,4.3.10),

3.2.2 Corrosion-resisting metal. Corrosion-resisting metal shall be corrosion-resisting steel conforming to ASTM A 269, grade TP 304, ASTM A 240, grade 304, or nickel-copper alloy conforming to ASTM B 127,

3.2.3 Pining. Steam and condensate lines shall be made of ASTM A 269 corrosion-resisting steel grade TP 304, or of copper in accordance with ASTM B 75.

3 2 4 Valves. steam and air. Steam and air valves shall be made of cast bronze in accordance with ASTM B 61. The trim (stem, disc, and seat) shall be made of corrosion-resisting steel, Valves shall be suitable for temperatures and pressures as specified herein.

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3.2,5 Fittings. pipe Pipe fittings shall be in accordance with ASTM B 61 or ASTM B 62.

3,2,6 Flexible steam hoses. Steam hoses shall be as specified in 3.2.6.1 through 3.2.6.5.

3.2.6.1 Pressure carrier. Steam pressure carriers shall be made of cold rolled steel in accordance with ASTM A 109.

3.2.6.2 Guard. The guard encasing around the wire braid shall be made of galvanized steel in accordance with ASTM A 446.

3,2.6.3 Wire braid The wire braid shall be made of stainless steel in accordance with ASTM A 478.

3,2.6.4 Coupling. Coupling shall be made of malleable iron in accordance with ASTM A 47.

3.2.6,5 Nipples. Nipples shall be made of malleable iron in accordance with ASTM A 47.

3,2.7 Conduit and wire. Electrical conduit shall be either rigid steel, zinc-coated conduit or electrical metallic tubing Wire shall be stranded annesled copper wire, sized properly for current loading, insulated to withstand press heat flux, and shielded

3,2.8 Type I - presses. Type I presses shall be as specified in 3.2.8.1 through 3.2.8.8.

3.2.8.1 Head and buck. The head and buck shall be constructed of malleable iron as specified in ASTM A 47 or steel as specified in ASTM A 36. The head and buck shall conform to the ASME Boiler and Pressure Vessel Code, Section VIII.

3.2.8.2 Spray chamber. The spray chamber shall be made of galvanized ferrous metal, corrosion-resisting metal plate, or aluminum alloy as specified in ASTM B 209,

3.2 8 3 Head operating handle. The head operating handle shall be made of molded plastic supported on metal brackets.

3.2.8.4 Head hose, (See 3.2.6,)

3 2.8.5 Head Press plate, The head press plate shall be made of aluminum and covered with a nylon-polyester cloth to prevent shining of garments

3 2 8 6 Buck covering. The buck shall be provided with a screen of copper, brass, stainless steel, or aluminum When copper, brass, or stainless steel is used, it shall be galvanically insulated from the aluminum castings Two pieces of 5/8-inch knitted cotton padding, one thickness of table felt (minimum weight-15 ounces per square yard) and an easily detached cover cloth shall be provided. Springs with hooks shall be provided to ensure an even tension to the buck covering for a smooth resilient pressing surface.

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3.2.8.7 Foot pedals. Foot pedals shall be fabricated of malleable iron, ductile iron, or steel

3.2.8.8 Table. The table shall be made of ferrous metal finished to prevent deterioration in a salty atmosphere (see 4.4.3.9).

3.2.9 Type III - spotting boards. Type 111 spotting boards shall be as specified in 3.2,9.1 through 3.2,9.5.

3.2.9.1 Spottiong board. The spotting board shall have a stainless steel frame supporting a smooth, stainless steel, vitrolite, or other material top that shall not allow spotting chemical stains and chemical deterioration (see 4.4.3 3) The narrow end of the board shall be made of perforated stainless steel with a fine nickel screen covering or a perforated teflon-coated nose covering.

3.2.9.2 Stand. The stand shall be made of ferrous metal materials finished to prevent deterioration in a salty atmosphere (see 4.4.3.9) .

3.2,9.3 Steam spotting gun The spotting gun shall be made of brass or corrosion-resisting metal (see 3.2.2), A heat insulated handgrip shall be of a material that will insulate the grip to ensure the surface temperature does not exceed 95 degrees Fahrenheit (“F), allowing comfortable operator use.

3.2 9.4 Swinging sleeve board. (See material requirements specified in 3.2.9.1.)

3,2.9.5 Garment tray. The tray shall have a ferrous metal frame and a canvas or other durable fabric cover attached to it.

3.2.10 Recovered materials, Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term “recovered materials” means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials, None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.2.11 Asbestos or asbestos-treated materials and cadmium plated materials Equipment specified herein shall be free from asbestos or asbestos-treated materials and cadmium plated materials.

3.3 Configuration. The configuration shall be as specified in 3,3.1 through 3 3 17 (see 6 3)

3 3.1 General. Dry cleaning presses and finishing equipment shall contain pneumatic, mechanical , and electrical systems, including motors, driving mechanism, starters, controllers, master switches, timers, reversing mechanisms, and other electrical equipments.

3 3.2 Tolerances. Tolerances shall be in accordance with ANSI Y14.5M.

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3.3.3 Fastener system. The extent and spacing of welds, rivets, bolts, and screws shall provide a durable construction that will withstand shipboard vibration (see 3.6.4) and prevent bulging of fastened parts.

3.3.4 Heating. mechanism. Equipment requiring heat shall be heated with 100 pounds per square inch gauge (lb/in^2) saturated steam.

3.3.5 Bearings. Contacting metal surfaces subject to movement during normal operation shall be provided with bronze bushings, ball, or roller bearings. All bronze bushings shall be oilless or oil impregnated.

3.3.5.1 Replaceability of bearing. Bearings shall be replaceable (i.e., bearings not enclosed within the equipment structure to prevent access for removal, bearings not welded to the equipment structure, special tools not required to gain access and/or removal, and similar) (see 3.3.17).

3.3.6 Lubrication. Equipment having moving parts shall be provided with fittings for lubrication and shall be accessible for maintenance equipment like; grease guns, oil funnels. oil cans, and similar items; in accordance with the requirements of 3.3.17. Except for motors and where lubricant reservoirs are required, forced lubrication with fittings shall be provided. Lubricant reservoirs shall be provided with a means of checking and maintaining lube oil or grease levels. Renewable oils and greases shall be of the following types:

(a) High-performance, multipurpose grease in accordance with DOD-G-24508. (This grease is intended for ball or roller bearings operating continuously up to 300°F.)
(b) Lubricating oil in accordance with MIL-L-17331, military symbol 2190 temperature extra pressure. (This oil is intended for general mechanical lubrication.)
(c) High temperature, lubricating grease (electric motor, ball and roller bearings) in accordance with MIL-L-15719. (This grease is intended for use in class M (silicone) electric motors operating above 300°F.)
(d) Silicone compound in accordance with MIL-S-8660. (This compound is intended to lubricate rubber.)

3.3.6.1 Seals. Seals shall be provided to prevent lubricants from entering parts of the machines where clothes are handled.

3.3.7 Mounting. Types I, III, and V equipment shall be deck-mounted.

3 3.7.1 Machine bases Bases shall be flat on the bottom. Lugs for anchor bolts shall be flush with the bottom of the base, or slightly relieved from the base, so that each unit can be installed on a shipboard foundation.

3.3.7.2 Machine mounting. Mounting lugs for type I units shall be configured as shown on figure 1. Other type units shall provide bolts to anchor units to the deck plate, No other attachment points shall be permitted except for ventilation, steam lines, air lines, water lines, and electrical cabling. Mounting lugs shall be attached to a sufficient area of the cabinet to ensure adherence under shipboard motion and vibration (see 3 6.4). The lugs, legs, and mounting frames shall be constructed not to exceed the yield point of its

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material when subjected to shipboard forces of 24 and 60 times the acceleration due to gravity in the horizontal and vertical directions, respectively, about the center of gravity of the unit (parallel to the front to back and side to side). Resilient mountings or springs shall not be used.

3.3.8 Piping Necessary integral piping, fittings, condensate traps, pressure-reducing valves and all other valves for air, steam, and drains shall be provided complete for installation to supply and drain lines. Interconnecting piping between equipments, and to connect equipment with supply and drain lines, shall be provided by the installing activity, Piping within and throughout the equipment shall be constructed to accommodate self- and ship-induced vibrations (see 3,6.4) without cracking, fatiguing, or rupturing.

3,3.8.1 Pipe connections to aluminum or aluminum-alloys, Where aluminum or aluminum-alloy is used in presses, pipe connections shall be fitted with reducing ferrous-metal bushings.

3.3.9 Controls, gauges , indicator lights. and audible signals Controls, gauges , indicator lights, and audible signals shall be as specified in 3.3.9 1 through 3.3.9 1.13

3.3.9 1 General criteria. Controls, gauges, indicator lights, and audible signals shall be selected and integrated into the dry cleaning presses and finishing equipment in order to meet the human factors engineering requirements specified in MIL-STD-1472 and herein

3.3.9.1.1 Distribution of work load. Controls shall be selected and distributed so that none of the operator's limbs will be overburdened Equipment design shall take into account the following considerations:

(a) The nature, frequency, and difficulty of the related tasks to be performed by the operator.
(b) The position of the body during the performance of tasks.
(c) Mobility or flexibility requirements imposed by these tasks

For further information on anthropometric data see MIL-STD-1472.

3.3,9.1.2 stops. Controls which should not be operated past their start/ end positions or limits shall be provided with stops to prevent such movement,

3.3.9.1.3 Consistency of movement. Direction of control movement shall be consistent with the related movement of an associated display or equipment component. Movement of a control forward, clockwise, to the right, up, or pressing a control, shall turn the equipment or component on, or cause the quantity to increase.

3.3.9.1.4 Grouping. Controls that function in sequential operation necessary to a particular task or that operate together, shall be grouped together along with their associated displays. When several steps of a sequence are selected by one control, the steps shall be arranged by order of occurrence to minimize control movements and prevent cycling through unnecessary steps Cycling through the control's ON and OFF position shall be avoided.

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3.3.9.1.5 Sequential operation. Where sequential operations follow a fixed pattern, controls shall be arranged to facilitate operation in a pattern left to right and top to bottom, as a printed page.

3.3.9.1.6 Location of primary controls. The most important and frequently used controls shall have the most favorable position with respect to ease of reaching and grasping (particularly rotary controls and those requiring fine settings) . When a trans-illuminated indicator is associated with a control, the indicating light shall be located so that it can be associated with the control without error and shall be visible to the operator during control operation.

3.3.9,1.7 Spacing. Minimum spacing between controls shall be in accordance with MIL-STD-1472. Spacing between a control and any adjacent obstruction shall be as specified in MIL-STD-1472. Spacing between adjacent edges of simple round indicator light fixtures shall be sufficient to permit clear labeling, signal interpretation, and bulb removal.

3,3.9.1.8 Control colors. Controls shall be black or grey. If color coding is required, only the following colors shall be selected for color coding: red, green, orange-yellow, white, blue. Blue shall only be used if an additional color is absolutely necessary.

3.3.9.1.9 Relation to display. When color coding must be used to relate a control to its corresponding display, the same color shall be used for both control and display. Color coding shall be in accordance with MIL-STD-1472

3.3,9.1.10 Control panel contrast. The color of the control shall provide contrast between the panel background and the control.

3.3.9.1.11 Coding. Indicator lights and controls shall be coded to conform to MIL-STD-1472. However, the different sizes specified in MIL-STD-1472 are intended only for the attention-getting value that larger lights of at least equal luminance provide in comparison to smaller, less noticeable indicator lights of lesser importance. No more than three different size lights and controls shall be used for discriminating importance by size.

3.3.9.1.12 Prevention of accidental actuation. Controls shall be constructed and located so that they are not susceptible to being moved accidentally. Controls shall be provided with resistance features such as viscous or coulomb friction, spring-loading, or guards.

3.3.9.1 13 Labelling of controls. Controls and displays shall be clearly labelled with plates containing the basic information needed for proper identi-fication, utilization, actuation, or q anipulation of the element, except where it is obvious to the operator what an item is and what is to be done with it. Labels shall convey verbal meaning in the most direct manner by using simple words or phases, Each control and display shall be labelled according to function, and the following criteria shall apply:

(a) Similar names for different controls and displays shall be avoided.
(b) Instruments shall be labelled in terms of what is being measured or controlled, taking into account the user and purpose.

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(c) Control labelling shall indicate the functional result of control movement (such as on and off). Such information shall be visible during normal operation of the control.
(d) Labels shall normally be placed above the controls and displays they describe, and shall be oriented horizontally so that they may be read from left to right. When the panel is above eye level, labels may be located below if the label visibility will be enhanced.
(e) The units of measurement (such as 'F, minutes, and so forth) shall be labelled on the panel.
(f) Lettering shall be dark on a light background,

3.3.9.1.14 Handwheel or crank. When a handwheel or crank is used to adjust the pressure between the press head and buck, MIL-STD-1472 shall be used as a guide .

3.3.9.1.15 Valves. Steam, air, and other operating valves shall be installed so that the valves can be operated from the front of the equipment. Rotary valve controls shall open the valve with a counterclockwise motion. Valve controls shall be provided with double-ended arrows showing the direction of operations and labelled at each end to indicate functional result (such as open and close) . Valves shall be readily accessible for maintenance (see 3.3.17). Globe or angle valves shall be equipped with a renewable corrosion-resisting or composition disk.

3.3.10 Vacuum pump, Each equipment item which requires vacuum shall be delivered with a 440-volt, 60-hertz (Hz), three-phase vacuum pump for operation with the type I and III equipment items ordered. The vacuum pump shall be designed in accordance with 00-D-735 (item 1, class 2) and amendments thereto.

3.3.11 Parts access. Guards shall be fitted over parts requiring frequent adjustment to permit access to such parts without requiring the use of special tools or inordinate disassembly and reassembly (see 3.3.17).

3.3.12 Cleaning access. Presses shall be constructed and located to permit cleaning underneath the machine when required Minimum access requirements shall be in accordance with MIL-STD-1472

3.3.13 Flexible steam hoses. Steam hoses shall be as specified in 3.3.13.1 through 3.3.13 4.

3.3 13.1 Pressure carrier. The pressure carrier shall be a closed pitch, corrugated, one-ply or multi-ply, and shall have annular or helical corrugations.

3 3 1 3 2 Braid reinforcement The braid reinforcement shall be secured by the end couplings and shall not have any loose wires protruding that could interfere with the tightness of the couplings,

3.3.13.3 Guard. Each hose shall have a protective wire guard helically wound outside the braid. The guard shall be attached by mechanical means, brazing, or welding. Pitch and spacing of the guard shall provide maximum protection to reinforce the braid, while not adversely affecting the flexibility of the hose

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3.3.13,4 Couplings, The hose shall include a coupling on each end of the hose securely fastened by brazing or welding. The hose couplings shall be appropriately sized to couple with the type I equipment steam fittings on the steam supply and return parts on the appropriate head, buck, and press locations.

3.3.14 Disassembly. Type I, classes B and C, and type V presses shall be constructed to be disassembled, transported, and reassembled within the ship. Major subassemblies that bolt together shall be preferred. Limiting dimensions for each major subassembly shall be 50 inches high, by 25 inches wide, by 36 inches deep. Subassemblies shall be match-marked and accompanied with adequate illustrations and directions to assist in field assembly.

3.3.15 Volumetric dimensions. The pressing and finishing equipment shall not exceed the dimensional limits specified in table I. Volumetric measurements shall be measured when the machine is in an open state.

TABLE I. Maximum volumetric dimensions of equipment

Type	Class	Long (inches)	Wide (inches)	High (inches)
I	B	60	44	63
I	C	66	44	63
III	-	50	24	50
V	A	43	28	76

3.3.16 Cover cloths and pads. If required, cover cloths and pads shall be provided complete with each item.

3.3.17 Human factors engineering Human factors engineering criteria, principles, and practices, as specified in MIL-STD-1472, shall be used in the design of the dry cleaning presses and finishing equipment where specified herein. The equipment shall be constructed so that access for operation, maintenance , and repair shall be analysed as specified in MIL-STD-1472. Access to each control, lubrication fittings, or any other component shall include space for any tools, protective clothing to be worn by the maintainer, or test equipment required, Maintenance and operational activities shall permit performance by the fifth percentile female to the ninety-fifth percentile male as specified in MIL-STD-1472. Particular attention shall be given to the ease of access for the maintenance of valves and replacement of bearings. Skill level and training requirements for operation, maintenance, and repair as influenced by the machine's design shall be kept at the lowest level consistent with other requirements specified therein. The analysis of operation, maintenance and repair functions shall reflect safety considerations

3 4 Construction. Construction shall be as specified in 3.4.1 and 3.4.2

3.4.1 General. Dry cleaning presses and finishing equipment shall be new and of rigid construction.

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3.4.2 Metal joints. Rivets, bolts, machine screws, nuts, and washers shall be made of steel, except when brass or corrosion-resisting metal is fastened, in which case they shall be made of brass or corrosion-resisting metal, respectively, Where dissimilar metals are fastened, rivets, bolts, machine screws, nuts, and washers shall be galvanically compatible with materials being joined. Electrodes or welding rods used for welding shall deposit material similar to the base metal . Nuts subject to vibration shall be adequately locked. The use of drill-point screws shall be prohibited. Bolts shall be used where practical, and blind joints shall utilize threaded inserts such as clipnuts, TSN nutserts, rivnuts, and so forth. Sheet metal self-tapping screws may be used for mounting small components (access panels, relays, and so forth), but unified fasteners shall be preferred.

3.5 Details of units or parts. Units or parts shall be as specified in 3.5.1 through 3.5.4.4.

3.5.1 Type I-presses, wool, utility. Type I presses shall be as specified in 3.5.1.1 through 3.5.1.14.4.

3.5 1.1 General. Each press shall consist of a rugged frame supporting a steam-heated head (or upper buck) conforming to a steam-heated lower buck. The frame shall support the work table and operating mechanism. The enclosure for the operating mechanism shall be provided with suitable panels (see 3.3.17) on the front and both sides that can be readily removed for inspection and maintenance. The base of each press shall be as shown in figure 1 for class B and C presses. Classes and sizes of type I presses shall be as specified in table II.

TABLE II. Classes and sizes of type I presses.

Class	Pressing surface		
	Length	Width of wide end	Width of narrow end
B	41 to 43 inches	12-3/4 inches	9-3/4 inches
C	45 to 47 inches	20 inches	9 inches

3.5.1.2 Press Pressure. Presses shall exert a minimum pressure force of approximately 3 lb/in^2 between the head and buck. The force shall be subject to variation by the use of a handwheel or some other device at the discretion of the operator. Presses shall be provided for use with air vacuum,

3.5.1,3 Drying device. The vacuum produced by the central system or to the individual attachment shall draw surplus steam and moisture from the garments through the lower buck after pressing. The control of this device shall be foot-operated (see 3,5.1.10), and the connection shall be made at the buck spray chamber.

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3.5.1,4 Vacuum valve The pipe to the vacuum source shall normally be closed to prevent the loss of spray steam, but shall be equipped with a valve that opens when the suction device pedal is depressed to apply vacuum.

3 5.1.5 Head and buck arrangement, The head, or both head and buck, shall be movable and spring-counterbalanced for ease of movement. The yoke shall be fitted with ball, roller, or bronze bearings, and shall be arranged so that it will not interfere with the placing of the goods or making proper contact with the opposite member when it is raised or lowered.

3.5.1.6 Head and buck adjustment. Springs, or other means, shall be provided for automatically adjusting either the head or buck, or both head and buck, from end to end and front to back to conform to varying thicknesses of goods being pressed

3.5.1 7 Steam and condensate. The head and buck shall be uniformly heated and properly drained The steam and condensate lines to and from moving members shall be connected by means of flexible metallic hose intended for steam working pressure of 100 lb/in^2.

3.5.1.8 Head. The head shall be concave in shape, conforming to the buck. with a pressure chamber for heating, and a spray chamber for moistening purposes. The spray chamber shall be formed by attaching a perforated metal plate to the concave surface of the head, or shall be cast integrally with, and supported by, the pressure chamber through numerous bosses and ribs. The rib and plate shall be removable. Press head shall conform to ASME Boiler and Pressure Vessel Code, Section VIII.

3.5.1 .8.1 Head pressing surface. The pressing surface of the head spray surface shall be perforated uniformly over its face.

3 5 1.8 2 Head steam valve. A packless head steam valve shall be provided to deliver steam to the spray chamber when the operator opens the valve.

3 5.1.8,3 Head operating handle. The head shall be equipped with an operating handle supported on metal brackets away from the head. The handle shall be constructed to prevent contact of the operator's hands with the heated metal brackets or head surface. The handle and surrounding area shall be constructed to remain comfortable for the operator and shall not exceed 95°F The head valve shall be readily operable while the operator's hand is on the handle of the head, The head handle shall be operable when the head of the press is outfitted with a ventilation hood.

3 5 1 8 4 Head hose Steam lines shall be located, insulated, and guarded to prevent the operator or maintenance personnel from incidental contact that might result in burn accidents (see 3 3.13). Steam lines shall be connected to the head by means of a metallic, flexible hose

3,5.1,8.5 Head overlap. The head shall overlap the buck all around by 1/2 inch or more to prevent the garments from being marked at the outer edge of the head and buck while being pressed.

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3.5.1.8.6 Head release, Means shall be provided to hold the head in the closed position until the operator desires to release it.

3.5.1.8.7 Head shock absorber. A shock absorber shall be provide d to permit checking of the head to a smooth, silent stop, but not to interfere with the rapid opening of the head.

3.5.1,8.8 Head uress plate . A press plate shall be provided for the head. The pressing surface shall be sufficiently knurled to prevent shine or gloss on woolens being pressed.

3.5.1.8.9 Head carrying member. The head shall be attached to a swinging or rocking member provided with a bearing at the fulcrum point. The head-carrying member shall be arranged so that the head can be raised to a position which will not interfere with the placing of work on the buck. The front of the head shall be not less than 14-1/2 inches from the buck in full open position.

3.5.1.9 Buck. The buck shall be convex in shape with a pressure chamber for heating and a spray chamber for moistening The spray chamber shall be formed by attaching a perforated plate to the buck that contacts the garment, or shall be integral with, secured to, and supported by the pressure chamber through numerous bosses and ribs. Press buck shall conform to the ASME Boiler and Pressure Vessel Code, Section VIII.

3. 5.1.9.1 Buck pressing surface . The buck pressing surface of the spray chamber shall be perforated uniformly over its face. The bosses shall provide the required heat conductivity.

3.5.1.9.2 Buck support. The buck shall be mounted on a support connected to the frame and shall have steam and drain connections. The top of the pressing surface shall be 40-1/2 inches plus or minus 1 inch from the deck.

3.5,1.9.3 Buck steam valve. A pedal-operated, packless buck steam valve shall be provided to deliver steam that is substantially free from entrained moisture to the spray chamber, when the operator opens the valve. The valve shall be arranged to afford uniform steam distribution over the entire pressing surface.

3.5.1.9 4 Buck spacing. The clearance between the buck and backboard shall be not less than 6 inches at the widest point of the buck.

3.5.1,10 Foot pedals. Foot pedals shall be constructed for use in accordance with MIL-STD-1472, and shall be fitted to the front of the frame with the length of travel not exceeding 4 inches.

3.5 1.11 Table. A rigid metal table with backboard and apron shall be provided. The table shall be sized to retain clothing overhanging the lateral edges of the buck to provide support to the garment during pressing operations and to prevent garment contact with the operating mechanisms, piping, and frame of the press. Corners shall be rounded or knuckled, and edges shall be rounded to preclude damage to garments or flatwork, Table material shall be corrosion-resisting steel or painted medium steel. The working surface of the table shall be located approximately 32 inches above the base plane of the press

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3.5.1.12 Pins. Pins subject to wear shall be hardened, smooth, and accurately ground to size

3,5.1.13 Utility hook-ups, The datum point is located at the deckplace about the front left bolt center, with the X axis running parallel to the front edge of the machine going to the right, Y axis running parallel to the left edge of the machine going back, and the Z axis pointing up.

i: unit vector for X-AXIS
j: unit vector for Y-AXIS
k: unit vector for Z-AXIS

(a) Type I, class B and C presses

(1) Steam inlet 1/2 inch female (FEM) national pipe thread (NPT) FEM NPT: location [18 inch i + 27-1/2 inch j + 27 inch k].
(2) Steam outlet 1/2 inch FEM NPT: location [0 inch i + 27-1/2 inch j + 25-1/2 inch k] .
(3) Air inlet 3/8 inch FEM NPT: location [right rear corner].
(4) Vacuum inlet 1-1/2 inch FEM NFT: location [left rear corner]

3.5,1.14 Class B and C Presses. pneumatically operated. Class B and C presses shall be as specified in 3 5.1.14.1 through 3.5.1.14 4

3.5.1.14.1 Operation Pneumatically operated presses shall be power driven by compressed air. They shall be constructed to operate on air pressures of 70 to 100 lb/in^2.

3.5 1.14.2 Pneumatic power application. Presses shall be arranged so that the head, shall be moved automatically to the full open position when the power is off. The head, shall be moved continuously and uniformly by air power only in order to make contact with the opposite member. When in this position, air power shall apply the necessary pressure. The amount of pressure shall be adjustable The press head, or both head and buck, shall provide automatic take-up to compensate constantly for padding shrinkage. The press shall be arranged to ensure satisfactory pressure on garments of unequal thickness, as well as those of uniform thickness .

3 5.1.14.3 Press controls. Power for moving members shall be controlled by simultaneously energizing two recessed buttons not less than 10 inches apart, thereby requiring the use of both hands of the operator during closing and locking operations, Upon releasing one or both buttons before the head has reached the closed position, the moving member shall promptly and automatically return to the open position, After the pressing operation, the activation of one or two buttons. or other manual control device, shall return the moving members to the open position Personnel between the fifth and ninety-fifth percentiles of anthropometric dimensions shall activate the press without bending, leaning, or excessive reaching (see 3.3.17).

3.5.1.14.4 Safety release. In the event of malfunctioning of controls, provision shall be made for readily releasing the head manually.

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3.5.2 Type 111 board, spotting steam unit. Type 111 steam unit shall be as specified in 3.5.2,1 through 3.5.2.8.

3.5.2.1 General. The steam spotting board shall be a unit constructed for removing spots. The unit shall consist of a spotting board, swinging sleeve board, steam spotting gun, stand, controls, garment tray, and shall have space for storing spotting compounds. Width and length of complete unit shall not exceed 21 by 50 inches, respectively. Table height shall be approximately 37 to 44 inches. The spotting board utilities shall include compressed air, wet and dry steam, and air vacuum.

3.5.2.2 Spotting board. The spotting board shall have a frame supporting a top. The narrow end of the spotting board shall be a perforated cover for use with the steam gun. The wide end of the board shall be used for tamping, and shall be securely fastened to a metal stand. The perforated area shall allow disassembly to remove excess chemicals and dyes. The vacuum shall provide continuous withdrawal of any moisture through the perforated end of the board.

3.5.2.3 Stand. The stand shall be the support for the spotting board, the swinging sleeve board, the spotting gun, controls, garment tray, and spotting compounds. The stand shall have anchor bolt holes for fastening the unit to the deck

3.5.2.4 Steam spotting gun. The steam spotting gun shall consist of a steam separator, valve and control, spring arm, gun, steam hose, and fittings. The steam gun shall be arranged for ready use on any part of the board and equipped for supplying either dry steam, wet steam, or compressed air, by operator use of a foot treadle. The gun shall have a heat-resisting handgrip,

3.5.2.5 Swinging sleeve board. The swinging sleeve board shall be of the same construction as the spotting board, but of a size for use with tight-fitting sleeves and other items not compatible to the spotting board. The swivel shall be arranged to permit swinging the arm out of the operator’s way, and so constructed that it may be secured when not in use.

3.5,2.6 Controls. The controls shall be operated by pedals constructed for use in accordance with MIL-STD-1472. Controls provide compressed air, wet and dry steam, and an air vacuum at the desire of the operator.

3.5.2.7 Garment tray. A garment tray shall be provided under the spotting board for supporting the garments being processed. This support shall consist of a frame attached at one end to the stand and a durable fabric cover attached to the frame. The garment tray size shall be not less than the spotting board size

3.5.2.8 Storage space. Space shall be provided at the wide end of the board for storing special spotting chemical compounds, As a minimum, the storage space shall allow four 4-1/2 inch diameter by 6-inch tall containers to be adequately stored and readily accessible.

3.5.3 Type V - presses. wool, topper, pneumatic . automatic. Type V presses shall be as specified in 3.5.3.1 through 3.5 3.9.

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3.5,3.1 Size of pressing surface for type V presses. The pressing surface shall handle waist sizes ranging from 26 to 48 inches.

3.5.3.2 Controls and adjustments Controls and adjustments shall be as specified in 3.5.3.2.1 through 3.5.3.216, inclusive.

3.5.3.2.1 Manual on-and-off switch. A manual on-and-off switch shall be provided to electrically energize the machine. An indicator light shall be visible to the operator to indicate energization of the press.

3.5.3.2.2 Manual steam foot switch. With the activation of a foot switch, steam shall be delivered to precondition the garment as long as the switch is depressed.

3.5.3.2.3 Manual air foot switch. With the activation of a foot switch, air shall be delivered to thoroughly dry the garment as long as the switch is depressed.

3.5.3.2.4 Automatic cycle, The automatic cycle shall be activated by a foot pedal. Separate adjustable O to 30 second timers shall be provided for automatic steam and air. Timers shall provide a means for recalibration of the timer knob.

3.5.3.2.5 Bag pressure control. A device shall be provided to adjust the pressure in the bag to allow adequate pressing of wrinkles from the garment.

3.5.3.2.6 Steam disbursement. A device shall be provided to adjust the amount of sprayed steam to precondition the pressing garment fibers.

3 5.3.3 Air bag The air bag shall be of a material that allows steam and air to adequately condition the garment being pressed. The air bag shall be readily removable for washing or replacement.

3.5.3.4 Condensate collector. The condensate collector shall prevent moisture from collecting and dropping into the garments being pressed.

3.5.3.5 Water gun mist, The water gun shall deliver a fine mist of water at the operator's demand. A cut-off valve shall be upstream on the waterline that shall be readily accessible to the operator (see 3.3.17),

3.5.3.6 Waistband holder. The waistband holder shall hold the waistband of the garment until the air bag is activated or manually deactivated.

3.5.3.7 Buck. Press buck shall conform to the ASME Boiler and Pressure Vessel Code, Section VIII. Buck shall be adequately constructed to heat the air to dry the garment being pressed,

3.5.3.8 Screen. A screen shall be provided for the blower air inlet to prevent the suction of lint and dirt into the air bag. The screen shall be readily removable for cleaning.

3,5.3,9 Electrical input. The press shall be 115 Vat, 60-Hz, single-phase electrical input,

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3.5.4 Electrical requirements. Electrical requirements shall be 115 Vat, 60-HZ, single-phase . All electrical equipment shall conform to the recommended practices for electrical installation on shipboard as specified in IEEE 45. Electrical equipment shall be constructed for operation on a type I electrical system as specified in MIL-STD-1399, section 300. Control equipment shall be dripproof to meet the test requirements specified in 3.6.10. Push buttons and switches shall be arranged for safe and convenient operation (see 3.3.17). Electrical equipment shall be constructed to operate satisfactorily and without hazard to personnel or equipment with an accidental ground on any conductor.

3.5.4.1 Motors, Motors shall be of dripproof construction and of a capacity to operate the equipment continuously at required speeds under load without exceeding allowable temperature current requirements of motor insulation. Motor bearings shall have adequate facilities for lubrication. Motors shall have proper starting and operating characteristics for the driven equipment. Motors shall have the following characteristics in accordance with MIL-M-17060-

Ambient temperature	104°F
Service	C
Enclosure	Dripproof, totally enclosed
Cooling	Natural
Horsepower	As required
Speed	Ac: constant
Duty	Continuous or cyclic
winding	Ac: (fractional horsepower) squirrel cage induction
Voltage	115 Vat, 60-Hz, single-phase
Bearings	Ball or sleeve
Insulation	Class B or F

3.5.4.2 Controllers, Controllers for power circuits shall switch all three lines in three-phase circuits, and both sides of single-phase circuits. Motor controllers, push buttons, and associated equipment shall be ac. Integral motor power circuits shall be three-phase and shall be controlled by contractors, which in turn may be activated by relays. Control circuits shall operate at 115 volts or less Solenoid valves shall be provided with high temperature epoxy encapsulated coils. The control equipment shall have the following characteristics,

Ambient temperature	104°F
Enclosure	Dripproof
Operation	Manual or magnetic
Type	Ac: across-line
Function	To suit equipment
Duty	To suit equipment
Performance	Manual nonautomatic Magnetic" semiautomati c
Protection	Thermal overload and low voltage Remote circuit for safety interlock

3 5.4.3 Switches and indicators. Each pneumatically and electrically operated pressing unit shall have limit switches, remote-control push-button switches. indicator lights, and gauges, as necessary, to perform the required

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functions. The push-button control station shall be mounted at points convenient to the operators, Push buttons shall be fully protected or recessed to prevent accidental operation.

3.5.4,4 Wiring methods and practices. Wiring shall be in accordance with MIL-E-917 and this specification. Wiring between different devices mounted on the machine shall be in rigid, oil-resistant metal conduit or liquid-tight, flexible metal conduit not smaller than l/2-inch electrical trade size. Fittings and junction box connections for conduit entrance shall be either threaded bushings or watertight, neoprene-bushed, and strain-relieved. Drainage holes shall be situated at the low point of all conduit runs or junctions. Wiring shall be with moisture-resistant thermolastic-insulated copper conductors rated for an operating temperature of 221°F. Power conductors shall be not smaller than no. 12 American Wire Gauge (AWG). Control conductors shall be not smaller than no. 14 AWG. Conductors shall be stranded. When wires are run through holes in partitions, the holes shall be equipped with grommets for mechanical protection of insulation. Care shall be exercised in running wires to ensure that they are not carried over or bent around any sharp corners or edges. Terminal blocks shall be plainly marked to correspond with markings on the electrical diagram. Wires run in conduit or in harness assemblies shall be marked every 12 inches to correspond with the electrical diagram, Solder lugs, soldered joints, and wire-nut type of connections shall not be permitted Electrical connections shall be made using solderless ring connectors. The ends of each wire shall be connected to terminals on the part or to the terminal boards by means of a pressure or solderless lug terminal. No more than three connections shall be made at each terminal. Equipment shall be used on an ungrounded electrical system and no grounds shall be permitted on the machine wiring. Connections shall be made between motors, controllers, timers, push buttons, and limit switches installed on the unit. Leads shall be extended from motors and control devices to a connection box and properly identified for connections to wiring beyond the unit, Machines with manual across-the-line starters, when mounted on the unit, shall be provided with wiring between the motor and starter. Electric cable length shall be kept to a minimum and shall have sufficient slack. Wiring within the control enclosure shall be neatly arranged into harnesses and secured with wire tie wraps. Cable clamps shall be used as necessary to locate harnesses and prevent movement that could cause rubbing or chaffing damage. The electrical controls shall be provided with a hinged access door with hardware to tightly hold it closed (key latches and magnetic latches shall not be allowed) which will provide a future means for preventative maintenance and repair. The access door shall be sized so that maintenance tasks shall be easily and safely completed by the fifth to ninety-fifth percentile person; and consideration for the use of tools or test equipment shall be taken into account (see 3.3.17).

3.6 Performance characteristics. Performance characteristics shall be as specified in 3 6 1 through 3.6.10.

3.6.1 Press operation. Dry cleaning pressing and finishing equipment provided for shipboard use shall have the same basic operating characteristics, as found commercially.

3 6,2 Working pressures. Working pressures shall be as specified in 3.6.2.1 and 3.6.2.2.

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3.6.2.1 Saturated steam. Pipes, fittings, coils, cylinders, chests, and chambers using steam shall have 100 lb/in^2 saturated steam working pressure (see 4.4.3.5 and 4,5.3,4.1).

3.6.2.2 Compressed air. Pipes, fittings, and all parts that are pneumatically operated shall have a minimum air operating pressure of 70 lb/in^2 gauge (see 4.4.3.6 and 4.5.3.4.2).

3.6.3 Shipboard inclined operation. Equipment shall operate as specified when the ship is rolling 12 degrees starboard in a 4-1/2 second period (see 4.4,3.1).

3.6.4 Shipboard vibration, Equipment shall operate normally while subjected to vibration levels of 4 to 25 Hz in accordance with MIL-STD-167-1, type I environmental equipment. Acceptability shall be contingent upon the ability of the equipment to perform its function during and after testing. Minor damage or distortion shall be permitted during testing, provided it does not in anyway impair the equipment in performing its principle functions, Major failures shall be those failures which cause malfunction of the equipment or present a safety hazard to personnel. Non-repetitive failure of parts such as wiring or solenoid valves that may easily be replaced or repaired are generally considered minor failures. Major and minor failure classification shall be left to the test engineer (see 4.4.3.2).

3.6 5 Shipboard power. Electrically powered equipment shall be tested as specified in 4.4.3.7 for the following shipboard requirements:

Steady state voltage	Unit shall operate satisfactorily during each test condition
Power interruption	Unit shall return to normal operation without deterioration in performance when power is restored.
Leakage current	Unit under normal operation shall have a leakage current not greater than 5 milliamperes.
Power factor	The unit's overall power factor shall be within the range of 0.8 lagging to 0.95 leading under average voltage tolerance conditions .
Insulation resistance	The unit's electrical components shall have a minimum cold insulation of 10 megohms.

3.6.6 Flexible steam hose strength. Flexible steam hose shall exhibit the strength characteristics as specified in table III without any signs of deformation or destruction (see 4.4.3.8.1 and 4.4.3 8.2).

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TABLE III. Flexible steam hose strength.

Hose size dia (inches)	Inside bend dia (inches)	Hydrostatic proof pressure (lb/in^2 gauge)	Lifetime cycles
1/2	16	400	20,000
3/4	21	400	20,000
1	26	400	20,000
1-1/4	30	400	15,000
1-1/2	35	400	15,000
2	44	400	15,000

3.6.7 Salt spray fog. Equipment shall operate satisfactorily in a salt spray environment; painted surfaces shall not show signs of blistering as a result of electrolysis; and other exposed surfaces shall not show signs of pitting, corrosion damage, or permanent damage (see 4.4.3.9).

3.6.8 Electric motors. Electric motors shall adequately perform when subjected to maximum operational load The electric motor shall be within plus or minus 5 percent of its required voltage, and shall not exceed its maximum rated current draw (see 4.4.3.4).

3.6.9 Operation. Each equipment type and class shall be verified by satisfactorily performing all the operations specified (see 4 4.3.3)

3 6.10 Enclosures. Equipment with electrical control panels or electric motors shall operate satisfactorily and not accumulate water within the enclosure (see 4.4.3.10).

3.7 Identification plate. Identification plates shall be made for the machine and its major components: electric motors, and other parts. Identification plates shall be made of corrosion-resisting steel or brass, Characters shall be not less than 3/16 inch in height. Characters shall be stamped on the plate. Stamping shall be not less than 0,003 inch deep. The thickness of the plate shall be 0.03 inch minimum. Plate information shall remain permanent and legible for the life of the equipment.

3.7.1 Mounting. Each identification plate shall be securely affixed to each machine in a conspicuous place.

3.7.2 Identification plate data. Data marked on each identification plate shall include the following:

(a) Manufacturer's type or model number, serial number, contract or order number, and date of manufacture.
(b) Salient design features including size, voltage, current, and working pressures.
(c) Name of machine and stock number (the stock number will be provided to the contractor),

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(d) Component identification (CID) number. The CID number entry shall consist of the letters CID, followed by a nine-digit number, which will be provided upon receipt of provisioning technical documentation, No unit shall be shipped without the CID number, and its associate components shall also be recorded on the packing list.

3 8 Safety. Dry cleaning pressing and finishing equipment shall incorporate the applicable electrical, mechanical, and hot surfaces design safety requirements of MIL-STD-1472 in addition to those safety design requirements specified herein.

3.9 Workmanship. The finished end items shall be free from defects such as fractures, splits, punctures, tears, dents, creases, bows, miscasts, deterioration, or malformation and as follows:

(a) Welds shall be continuous (on structural supports), sound, smooth, and free from porosity, cracks, incomplete fusion, and deformation of material All scale and flux (when flux is used) shall be removed from the welds.
(b) Threaded fasteners shall not be broken, stripped, bent, fractured, loose , or missing,
(c) Rivets shall be driven completely to fill holes. The rivet heads shall be in full contact with the surfaces of the riveted member.
(d) Wiring shall not be cut, abraded, or have excess insulation stripped. Wiring shall be tightly joined at the terminals. Wiring harnasses shall be secured using clamps and/or tie-wraps to prevent chaffing, Wiring shall not be bent or carried through on sharp protruding edges.
(e) Surfaces shall be free from burrs, sharp edges, or slivers.
(f) Bearings, shafts, and gears shall be properly aligned.
(g) Safety guards shall be securely fastened to the equipment.
(h) Pressure vessels shall be ASME stamped and approved.
(i) The finish applied to the end item shall be continuous, smooth, adherent, without discolorations or foreign material embedded, and contains no sags, runs, drips, creeps, laps, bubbles, streaks, wrinkles, blisters, cracks, scratches, pores. pots, lumps, flux, or orange peel. No rust, rough grinds, or toolmarks shall show through the coatings.

4. QUALITY ASSURANCE PROVISIONS

4 1 Responsibility for inspection Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Pre-award survey.
(b) First article inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified. all inspections shall be performed in accordance with the test conditions specified in (applicable test method document or applicable paragraphs in the specification).

4.3.1 Intermediate visual and dimensional examination. The intermediate product or assemblies that cannot be checked or measured after final assembly shall be examined at various stages of manufacture for conformance to this specification. Examination shall be made at an appropriate time for defects in construction, workmanship, dimensions, and material. Any deviation from specified requirements shall be classified as a defect, Sampling shall comply with table IV

TABLE IV. Sampling for intermediate visual and dimensional examination

Lot size	Sample size	Accept	Reject 1/ 2/ 3/
2 - 15	2	0	1
16 - 25	3	0	1
26 - 90	5	0	1
91 - 150	6	0	1
151 - 280	7	0	1
281 - 500	9	0	1
501 - 1200	11	0	1
1201 - 3200	13	0	1
3201 - over	15	0	1

1/ All defective items must be replaced with acceptable items prior to lot acceptance
2/ Inspect sample size until reject criteria is reached.
3/ Rejected lots may be screened and resubmitted for inspection and retest.

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4.4 First article inspection. The first article inspection shall consist of the examinations shown in table V and the tests specified in 4.4.2 and 4.4,3.

TABLE V. Classification of defects,

Examine	Defect (requirement paragraphs)	Classification	
		Major	Minor
Materials	Construction materials, not as specified (see 3 2 1 through 3.2.11).	X	
Configuration	Configuration characteristics not as specified (see 3.3.1 through 3.3.17)	X	
Construction	Metal joints not as specified (see 3.4.2).		X
Details of type I presses	Size of pressing surface not as specified (see 3.5.1.1 and table II)	X	
	Panels not provided for inspection or maintenance as specified (see 3.5.1.1).	X	
	Deck mounting configuration not as specified (see 3.5.1.1).	X	
	Vacuum not attached to the buck spray chamber as specified (see 3.5.1.4).	X	
	Head and buck arrangement not as specified (see 3.5.1.5).	X	
	Head configuration and details not as specified (see 3.5.1.8).	X	
	Head is not approved as specified by ASME code (see 3.5 1.8).	X	
	Head pressing surface not as specified (see 3.5.1.8.1 and 3.5.1.8.8)	X	
	Head steam valve and handle not as specified (see 3.5.1.8.2 and 3.5.1.8.3).	X	
	Flexible steam line not as specified (see 3.5.1.8.4).	X	
	Head overlap not as specified (see 3.5 1.8.5).	X	
	Head carrying member not as specified (see 3.5 1.8.9).	X	

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TABLE V. Classification of defects - Continued.

Examine	Defect (requirement paragraphs)	Classification	
		Major	Minor
	Buck configuration and details not as specified (see 3.5.1.9).	x	
	Buck is not approved as specified by ASME code (see 3.5.1.9).	x	
	Buck pressing surface not as specified (see 3.5.1.9.1).	x	
	Buck steam valve and pedal not as specified (see 3.5.1.9.3).	x	
	Buck support not as specified (see 3.5.1.9.2).	x	
	Buck spacing from backboard not as specified (see 3,5.1.9 4).	x	
	Foot pedals not as specified (see 3 5 1 10)	x	
	Table not as specified (see 3.5.1.11).	x	
	Pins not as specified (see 3.5.1.12).	x	
	Utility connections not as specified (see 3.5,1,13).	x	
	Press controls not as specified (see 3.5.1. 14.3 and 3.5.1.14.4).	x	
Details of type III spotting board	Dimensions not within specified limits (see 3.5.2.1).	x	
	Utilities not as specified (see 3.5.2.1).	x	
	Spotting board not as specified (see 3 5.2.2).	x	
	Stand not as specified (see 3.5 2 3)	x	
	Steam spotting gun not as specified (see 3.5.2.4)	x	
	Swinging sleeve board not as specified (see 3.5.2.5).	x	

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TABLE V. Classification of defects - Continued.

Examine	Defect (requirement paragraphs)	Classification	
		Major	Minor
	Controls not as specified (see 3.5.2.6).	X	
	Garment tray not as specified (see 3.5.2.7).	X	
	Storage space not as specified (see 3.5.2.8).	X	
Details of type V press	Controls and adjustments not as specified (see 3.5.3.2 through 3.5.3.2.6).	X	
	Air bag not as specified (see 3.5.3.3).	X	
	Water gun with cut-off valve not as specified (see 3.5.3.5).	X	
	Buck not as specified (see 3.5.3.7).	X	
	Screen not as specified (see 3.5.3.8).	X	
	Electrical input not as specified (see 3.5.3.9).	X	
Electrical requirements	General requirements not as specified (see 3.5 4).	X	
	Motors not as specified (see 3.5.4.1).	X	
	Controllers not as specified (see 3.5.4.2).	X	
	Switches and indicators not as specified (see 3.5 4.3).	X	
	Wiring not as specified (see 3.5.4.4).	X	
Identification plate	Plate material not as specified (see 3.7).	X	
	Plate not mounted securely (see 3 7.1).	X	
	Plate data not as specified (see 3.7 2)	X	
Safety	Safety guards not as specified (see 3 8)	X	

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TABLE V. Classification of defects - Continued.

Examine	Defect (requirement paragraphs)	Classification	
		Major	Minor
Workmanship	General workmanship not as specified (see 3.10).	X	
	Welds not as specified (see 3.9(a)).		X
	Fasteners not as specified (see 3.9(b)).		X
	Rivets not as specified (see 3.9(c)).		X
	Wiring not as specified (see 3.9(d)).	X	
	Surfaces not as specified (see 3.9.(e)).		X
	Alignment of bearing, shafts, and gears not as specified (3.9.(f)).	X	
	Guards not as specified (3.9(g)).	X	
	Pressure vessels not as specified (3.9(h)).	X	
	Finish not as specified (3.9(i)).		X

4.4 1 First article sample. Prior to the start of production, one machine of each design and size shall be made ready for approval of the NAVSSES 032E representative by the contracting officer's authorized representative for the tests and examinations specified in 4.4,2 and 4.4.3. The contractor may utilize his own facilities, or any commercial laboratory acceptable to the Government. Approval of the sample by the NAVSSES 032E representative shall not relieve the contractor of the responsibility of conformance to this specification, Disposition instructions for the first article sample shall be obtained from the contracting officer's designated technical representative (see 6.2)

4 4 2 Contractor tests Before submission of the first article sample for inspection, the contractor shall subject the machine to the tests specified in 4.4.3. The contractor shall provide calculations for the mounting feet pads for the equipment and attached subassemblies when subjected to the loading specified in 3.3.7.2. The calculations shall include the shear stress and tensile stress for the hold-down bolts and feet pads, the shear stress on the welding in the localized area, and the bearing stress on the feet pads. The calculations shall be compared with the shear and tensile strength of the material undergoing the loaded conditions, When specified in the contract or order, a first article test report shall be prepared (see 6.3).

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4 4.3 Inspection tests. Inspection tests shall be as specified in 4,4.3.1 through 4.4.3.10.

4,4.3.1 Inclination. Each machine shall be inclined at an angle of 12 degrees to each side of two vertical planes at right angles to each other, and shall operate satisfactorily in each of the four positions for 30 minutes (see 3.6.3).

4,4.3.2 Vibration testing. Types I, III, and V machines shall be tested under normal operating conditions in accordance with MIL-STD-167-1. The equipment shall be tested at a frequency range of 4 to 25 Hz. Presses shall meet the requirements specified in 3.6,4.

4,4.3.3 Functional testing, Each machine shall be checked and tested to determine conformance to the following:

(a) Type J presses:

(1) Recessed two-button, hand/head closure buttons at least 10 inches apart.
(2) Automatic adjustment to varying thicknesses of goods be--rig processed.
(3) Non-locking head closure on any obstruction greater than 3/4 inch.
(4) Steam injection through garments using the head steam-operating handle and the buck pedal-operated steam valve.
(5) A vacuum system to draw surplus steam and moisture from garments through the lower buck when the pedal-operated vacuum valve is activated,
(6) Head release shall be by button, or by release lever bar.
(7) Press shall open when the air supply is turned off.
(8) Head and buck pressure shall be a minimum of 3 lb/in^2.
(9) Head closure, and opening to and from the buck, shall be smooth.
(10) Air pressure shall be between 70 and 100 lb/in^2 gauge,
(11) Each press class shall adequately press garments for a duration of 10 hours

(b) Type 111 spotting boards:

(1) The swinging sleeve board shall swivel above the spotting board and lock into position. Likewise, the sleeve board shall swivel away from the spotting board, out of the operator's way.
(2) The spotting gun shall provide dry steam, wet steam, or compressed air when the appropriate foot pedal is activated. It shall be demonstrated that a fine silk garment, when sprayed with dry steam at a distance of 4 inches for 5 seconds, shall show no signs of damage.
(3) The perforated area on the spotting board and sleeve board shall demonstrate moisture removal from garments, with the vacuum applied by foot pedal. It shall be easily disassembled to remove excess chemicals and elves.

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(4) The storage space provided at the wide end of the spotting board shall provide, as a minimum, the following:

(a) An area large enough to store four 4-1/2 inch diameter by 6-inch tall containers.
(b) Adequate space for easy access to these containers.
(c) Adequate means to securely hold these containers during rough seas.

(5) Spotting boards shall adequately remove stains from garments for a duration of 4 hours. The spotting board shall demonstrate it's ability to resist spotting chemical staining or deterioration.

(c) Type V presses:

(1) The on and off switch shall provide an indicator light to verify activation.
(2) Variable timers for steam and air shall be adjustable from 0 to 30 seconds and shall automatically reset after cycle completion.
(3) Activation of the automatic pressing cycle shall be by foot pedal. This activation shall provide steam to the buck surface co condition the garment, followed by warm air to remove the moisture from the garment for the time setting (maximum 30 seconds per mode) of the automatic cycle for each mode.
(4) The manual steam foot pedal shall deliver steam to the buck assembly when activated.
(5) The manual air foot pedal shall deliver warm air to the buck assembly when activated.
(6) A damper control shall adjust air pressure in bag,
(7) The water spray gun shall deliver a fine water mist,
(8) The expander plate shall hold the back part of the waistband until the bag expander is activated.
(9) The pant topper shall be worked for 10 hours and shall adequately finish pants, with and without pleats, ranging from 26 to 48 inches in waist sizes.

4.4.3.4 Power input, Motor shall not draw in excess of rated current after accelerating to full speed under its maximum loaded conditions. The voltage to the motor shall be measured to determine if it is within 5 percent of its rated value . Equipment shall meet the requirements specified in 3.6.8,

4 4.3 5 Steam assemblies. Steam assemblies shall be tested hydrostatically at 150 lb/in^2 gauge (which is 150 percent of the maximum system operating pressure) for 30 minutes with no indication of leaks or distortion, Steam assemblies shall meet the requirements specified in 3.6.2.1.

4.4.3.6 Pneumatic assemblies. Pneumatic assemblies shall be tested with 130 lb/in^2 gauge air (which is 150 percent of the maximum system operating pressure) for 30 minutes without indication of leaks or distortion, Pneumatic assemblies shall meet the requirements specified in 3.6,2,2.

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4 4.3 7 Electrical tests Equipment using electricity shall meet the applicable requirements specified in 3.6.5, and shall be tested as specified in 4 .4.3.7.1 through 4.4.3.7.5, inclusive.

4.4. 3.7,1 Steady state voltage and frequency. The equipment shall be operated for at least 15 minutes at the following conditions:

(a) 115 Vac supplied equipment

(1) 115 Vat, 60-Hz
(2) 109 Vat, 60-Hz
(3) 121 Vat, 60-Hz

4 4.3.7.2 Power interruption. With the equipment operating within the steady state voltage and frequency tolerances specified in 4,4.3.7.1, the external power supply shall be suddenly interrupted. After an interval of 0.5 to 20 seconds, the power supply shall be reapplied within the steady state tolerances , and the equipment restarted. The test shall be repeated four times

4 4.3.7.3 Leakage current. With the frame of the equipment ungrounded, and the equipment operating within the steady state voltage and frequency tolerances specified in 4,4.3,7 1, the leakage current for each phase between the frame shall be successively measured through a 1500-ohm resistive load.

.44.3 ,74 Power factor With the equipment operating within the steady state voltage and frequency tolerances specified in 4.4.3.7.1, the voltage, current , and wattage shall be measured, and the power factor computed as follows:

Single-phase:

$$\text{Power factor} = \frac{\text{Wattage}}{\text{Voltage}_{\text{average}} * \text{Current}_{\text{average}}}$$

4 4 3 7 5 Insulation resistance. The insulation resistance between all mutually insulated points, and between all insulated points and the frame, shall be measured with a 500 Vdc megohmmeter.

4.4.3,8 Flexible hose tests. Flexible hose tests shall be as specified in 4.4.3 8.1 and 4.4.3.8.2. Flexible hoses shall meet the requirements specified in 3.6.6

4.4.3.8.1 Flexible steam hose Pressure proof test. The hose shall be subjected to a proof pressure specified in table III for a duration of 15 minutes without signs of destruction. The hose shall be coiled into a circle of its bend diameter. The coil shall be rolled the entire length of the hose 10 times, while being subjected to a hydrostatic pressure of 200 lb/in^2 gauge. Finally, after successful completion of the preceding tests, the hose shall be tested at a minimum pressure of 10 lb/in^2 gauge air, while underwater, to detect signs of leakage.

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4,4,3.8.2 Life test for flexible steam hose. The hose shall be flexed to meet the minimum requirements specified in table III. The cycling rate shall be 20 per minute, with the hose subjected to 200 lb/in^2 gauge at 425°F steam service conditions . Hose shall be flexed to and from its bend diameter.

4.4.3.9 Salt spray fog. Equipment shall be exposed to a salt fog concentration of 5 ± 1 percent, with the unit set-up in its normal operating condition. The test item shall be conditioned in the chamber for 2 hours before the salt spray fog. The unit shall be exposed for a period of 48 hours, followed by a 48-hour drying period. The salt spray fog shall be emitted from an orifice with a diameter size of 0.02 to 0.03 inch that atomizes approximately 2.8 liters of salt solution per 10 cubic feet of chamber volume in a 24-hour period. The temperature of the chamber shall be maintained at 95° ± 2°F, and the salt spray fog shall be maintained within plus or minus 10°F of the chamber temperature. Equipment shall meet the requirements specified in 3.6.7,

4.4.3.10 Dripproof test. Equipment containing electrical controls or motors shall undergo the following dripproof test. One thousand cubic inches per square foot of area shall be covered by the drip or sprinkle in 60 minutes. The dripping or sprinkle rate and distribution shall be as uniform as possible. The enclosed equipment shall be oriented in five different positions during the test; normal position, and inclined at 15 degrees forward, backward, and to each side. The minimum time in each position shall be 12 minutes (8 minutes with the equipment operating, and 4 minutes off).

4.5 Quality conformance inspection Quality conformance inspection shall be as specified in 4.5.1 through 4.5.4, inclusive.

4.5.1 Inspection lot. A lot shall consist of all machines of one design offered for delivery at one time.

4 5.2 Sampling for visual and mechanical inspections. Visual and mechanical examination of the end item shall be for defects specified in table V. Sampling shall be in accordance with table VI

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TABLE VI. Sampling for visual and mechanical inspections.

Lot size	Sample size	Accept	Reject 1/ 2/ 3/
Major Defects			
2 - 150	20	0	1
151 - 200	32	1	2
201 - 500	50	2	3
501 - 1200	80	3	4
1201 - 3200	125	5	6
3201 - 10000	200	10	11
10001 - 35000	315	14	15
35001 - Over	500	21	22
Minor Defects			
2 - 15	3	0	1
16 - 25	5	0	1
26 - 50	8	1	2
51 - 90	13	2	3
91 - 150	20	3	4
151 - 500	32	5	6
501 - 1200	50	8	9
1201 - 3200	80	14	15
3201 - Over	125	21	22

1/ All defective items must be replaced with acceptable items prior to lot acceptance,
2/ Inspect sample size until reject criteria is reached.
3/ Rejected lots may be screened and resubmitted for inspection and retest,

4.5.3 Sampling for test requirements. Test machines shall be selected in accordance with inspection level S-3, and shall be tested as specified in 4,5.3,1 through 4.5.3.5, inclusive. Sampling shall be in accordance with table VII,

TABLE VII Sampling for test requirements

Lot size	Sample size	Accept	Reject 1/ 2/ 3/
2 - 25	3	0	1
26 - 50	5	0	1
51 - 90	6	0	1
91 - 150	7	0	1
151 - 280	10	0	1
281 - 500	11	0	1
501 - 1200	15	0	1
1201 - 3200	18	0	1
3201 - 10000	22	0	1
10001 - Over	29	0	1

See footnotes at top of next page.

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1/ All defective items must be replaced with acceptable items prior to lot acceptance.
2/ Inspect sample size until reject criteria is reached.
3/ Rejected lots may be screened and resubmitted for inspection and retest.

4.5.3.1 Power input. The power input shall be measured, and the motor temperature rise determined, where applicable, during the operational test of 4.5.3.3. Equipment shall meet the requirements specified in 3.5.4.

4.5.3.2 Insulation testing. An insulation resistance test shall be conducted on each test machine, Insulation resistance of motor, controller, and connecting cable shall be measured by a 500-Vdc insulation resistance tester. Insulation resistance for each component shall be not less than 10 megohms. Equipment shall meet the requirements specified in 3.6.5

4.5.3.3 Operational test. Each test item of laundry pressing and finishing equipment shall be connected to required utilities and operationally tested, without garments, to determine conformance to this specification. This test shall ensure that

(a) The assembly of component parts is complete and of proper inter-relationship ,
(b) There is no undue vibration, noise, or overheating.
(c) The safety devices operate properly, and the operation of the controls is smooth and positive.

4.5.3.3.1 Functional test. One item of each type and class under contract or order shall be tested by connecting to required utilities. The equipment shall operate for not less than 1 hour, with garments, to determine conformance to this specification, and to demonstrate that the item under test shall satisfactorily perform the function for which constructed

4.5.3.4.1 Leakage test (steamn and pneumatic). Leakage tests shall be as specified in 4.5.3.4 1 through 4.5.3.4.2, inclusive.

4.5.3.4.1 Steam. A hydrostatic test shall be performed on types I, III, and V equipment. Steam chambers shall be tested at 150 percent of their maximum rated working pressure for 30 minutes without indication of leads or distortion. After successful com letion of this hydrostatic test, the chambers shall be tested with 30 lb/in^2 air pressure for 1 minute to determine that there are no imperfections in castings, welds, or joints. A foaming aqueous solution shall be applied to all castings, welds, or joints to identify any imperfections that would cause leaks or equipment damage.

4 .5.3.4.2 Pneumatic (air). A hydrostatic test shall be made on types I and III equipment. Pneumatically-operated parts shall be tested at 150 percent of their maximum rated working pressure for 30 minutes without indication of leaks or distortion. After successful completion of this hydrostatic test, the chambers shall be tested with 30 lb/in^2 air pressure for 1 minute to determine that there are no imperfections in castings, welds, or joints. A foaming aqueous solution shall be applied to all castings, welds, and joints to identify any leakage.

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4.5.3.5 Dielectric test. Sample dry cleaning and finishing pressing equipment supplied with electricity shall be subjected for 1 minute to a dielectric test voltage, the effective potential of which is twice the rated voltage, plus 1,000 volts. This voltage shall be applied between points of polarity, and also between live parts and ground. The frequency of the test voltage shall be 60-Hz, rms, ac, and shall approximate a true sine wave. Equipment shall meet the requirements specified in 3.6.5.

4.5.4 Rejection. Failure of any test units to pass any of the tests of 4.5.3.1 through 4.5.3.5 shall be cause for rejection of the lot.

4.6 Test reports and certifications. When specified in the contract or order, reports, procedures, and certifications shall be prepared (see 6.3).

4.7 Inspection of packaging. Sample packages and packs, and the inspection of the preservation-stowage , packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5 PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging or preparation for delivery requirements of referenced documents listed in Section 2, see 6.7.)

5.1 Packaging requirements, Dry cleaning presses and finishing equipment shall be preserved level A or commercial, packed level A, B, or commercial, and marked for each level of protection in accordance with MIL-L-3153 including packaging acquisitioning option stated therein as specified (see 6.2). In addition, for Navy acquisitions, the following applies:

a Navy fire-retardant requirements.

(1) Lumber and plwood. When specified (see 6.2), all lumber and plywood including laminated veneer material used in shipping containers and pallet construction, members, blocking, bracing, and reinforcing shall be fire-retardant treated material conforming to MIL-L-19140 as follows:

Levels A and B - Type II - weather resistant.
Category I - general use.
Level C - Type I - non-weather resistant.
Category I - general use.

(2) Fiberboard, Unless otherwise specified (see 6.2), fiberboard used in the construction of class-domestic, non-weather resistant fiberboard, and cleated fiberboard boxes including interior packaging forms shall meet the flame spread index and specific optic density requirements of PPP-F-320 and amendments thereto.

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5.4 Technical Manuals. Two copies of the current operation/maintenance technical manual(s) shall be packaged with each system or equipment, Technical manual packaging shall be as specified (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Commercial dry cleaning pressing and finishing equipments acquired under this specification are intended for shipboard use. Therefore, commercial machines offered to the Government may require alteration to meet specification requirements such as mounting provisions, electrical power, or materials.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification.
(b) Type and class of item required (see 1.2).
(c) Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(d) When first article inspection is required (see 3,1),
(e) Disposition of first articles (see 4.4.1).
(f) Level of preservation, level of packing and packaging options required (see 5.1).
(g) When fire retardant treated lumber and plywood is required (see 5.1a(1)).
(h) When fire retardant fiberboard is not required (see 5.1a(2)).
(i) Technical manual packaging requirements (see 5.4).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/ provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423,

Reference parazraph	DID Number	DID Title	Suggested Tailoring
3.3	DI-E-7031	Drawings, engineering and associated lists	Level 3
4.4.2	DI-T-4902	First article inspection report	----
4.6	DI-T-2072	Test reports	----
4 6	DI-T-5329	Inspection and test reports	----
4.6	DI-E-2121	Certificate of compliance	10 .1

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The above DID's were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) , must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.4 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.4.1 Technical manuals development, Technical manuals (including technical repair standards (TRS)) should be procured under separate CLINs or subCLINs. Technical manual products (such as manuscript copy, preliminary copy, and reproducible copy) should be developed and delivered as specified in the Technical Manual Contract Requirements (TMCR) exhibit(s) to the contract The TMCR is an exhibit or attachment to a contract specifying technical manual product development and delivery requirements, and/or management data development requirements. CDRLs and DID's should not be used for technical manual product procurements. Technical manual management data (such as plans and reports) should be procured under a separate data CLIN, but delivery should continue to be specified on CDRLS and DID's. Technical manual quality assurance management data should be prepared and delivered as specified in the TMCR and contractor order.

6.5 First article, When a first article inspection is required, the item should be a first article sample. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.6 Provisioning. Provisioning Technical Documentation (PTD), spare parts and repair parts should be furnished as specified in the contract.

6.6.1 When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment Packaging for such parts should also be specified

6,7 Sub-contracted material and parts. The packaging or preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation, into the equipment and lose their separate identity when the equipment is shipped.

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6.8 Subject term (key word) listing

Pants topper
Pneumatic
Spotting board
Spray chamber
Steam pressure carrier
Swinging sleeve board

6.9 Changes from Previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 3510-N304)

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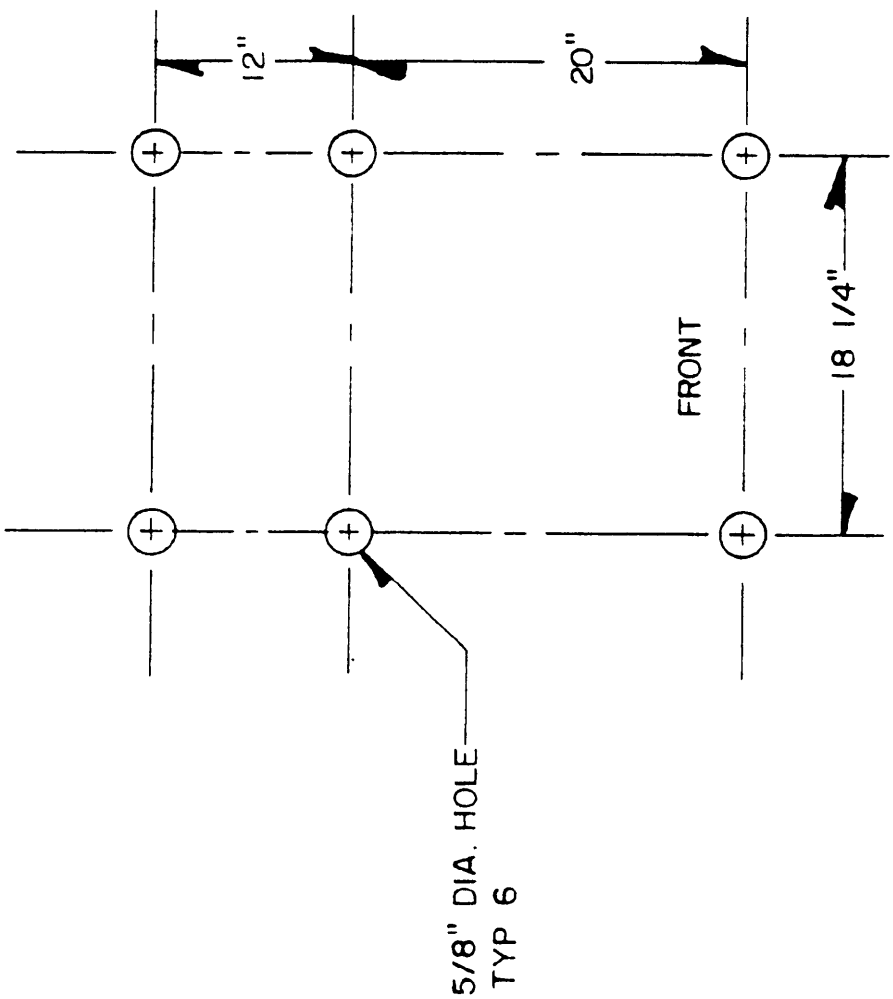


FIGURE 1. Type I, class B, C press foundation.

SH 1323229

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-D-22000B(SH)	DRY CLEANING PRESSES AND FINISHING EQUIPMENT, COMMERCIAL (SHIPBOARD)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a Paragraph Number and Wording

b Recommended Wording

c Reason/Rationale for Recommendation

6 REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE

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